

Evan Jones
U.S. Olympic & Paralympic Committee Practicum
Bi-weekly progress report
Week ending January 26th

Dear Mr. Schwartz,

Given that the period that this report covers is the first two weeks of the project majority of my time spent towards the project so far has been getting acquainted with both our project requirements and goals as well as with the company data. At your recommendation to focus on a few or select area/sporting event, I have done some research into the skiing events and their unique rules and structures. Although I haven't narrowed my scope completely yet, I'm fairly certain that this will be my primary area of focus for the project. From a data perspective, I also spent the past weeks conducting some exploratory data analysis. Most of this work centered on the visualization of histograms from various different sports showing the cumulative medal totals by age. Although a non-exhaustive search/analysis, after looking over several different sports and time-periods, it became clear that most sports have a distinct "peak" age where majority of all medals are accumulated. Interestingly, this distribution appears non-gaussian, and more strongly resembles a power distribution however this will have to be confirmed through additional investigation. My working approach for this project centers upon having reliable estimates of the historical medaling performance of athletes by age. Once I have a better understanding of this distribution, I can utilize simulation methods to approximate a given country's expected medal count given athlete ages. Although rather simplistic, this modeling approach offers high interpretability and gives a great starting place for subsequent modeling should it be insufficient with regard to out-of-sample performance. Over the course of the next two weeks, I plan to explore/apply some non-parametric analysis of these medal/age distributions primarily through usage of kernel density estimation. Additionally, I plan to take care of some of the infrastructure aspects of the project setting up an SQL database and implementing some various utility functions that will be needed for data related workflows later in the project. Thank for you continued help and support on this project. Let me know if you have any additional questions, and I will be happy to clarify.

Best regards,

Evan Jones
evanjones@gatech.edu
678-588-8666